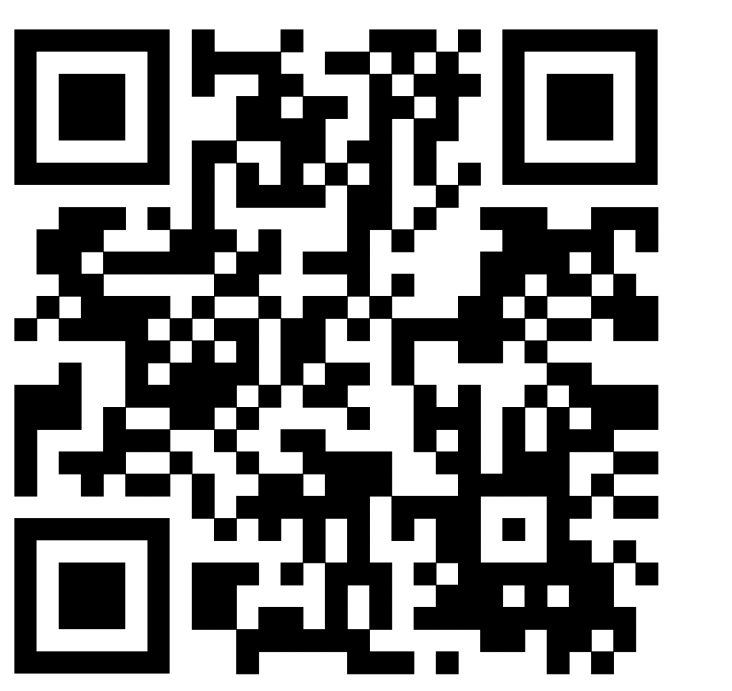




Two typical challenges of current SoTA attribution evaluation systems

• Challenge 1: Fine-grained information insensitivity (66.0% among all errors)

Claim: The population of Thailand is about 63 million people. [1]

References: [1] Thailand is home to about 63 million people, 95% of these people follow Theravada Buddhism and ... Thailand's war on drugs may potentially explain its current retention of the death penalty. The manufacture and distribution of drugs ...

GPT-3.5 (w/ CoT): ... The reference discusses the population of Thailand and its connection to Theravada Buddhism, the war... However, it does not provide specific information or data to support the claim that the population of Thailand is about 63 million people.

Final judgment: not attributable

Ground truth: attributable

• Challenge 2: Human-model accessible information mismatch (26.8% among all errors)

Claim: In delict, causation is typically divided into two components: causation in fact and legal causation. [4]

References: [1] In this section, causation in fact and legal causation are examined as well as situations where the defendant may be insulated from criminal responsibility. ... In this example, ...

GPT-3.5 (w/ CoT): ... The reference discusses the concept of causation in fact and legal causation in the context of criminal law. ... However, it does not explicitly mention the division of causation into two components..

Final judgment: not attributable

Ground truth: attributable

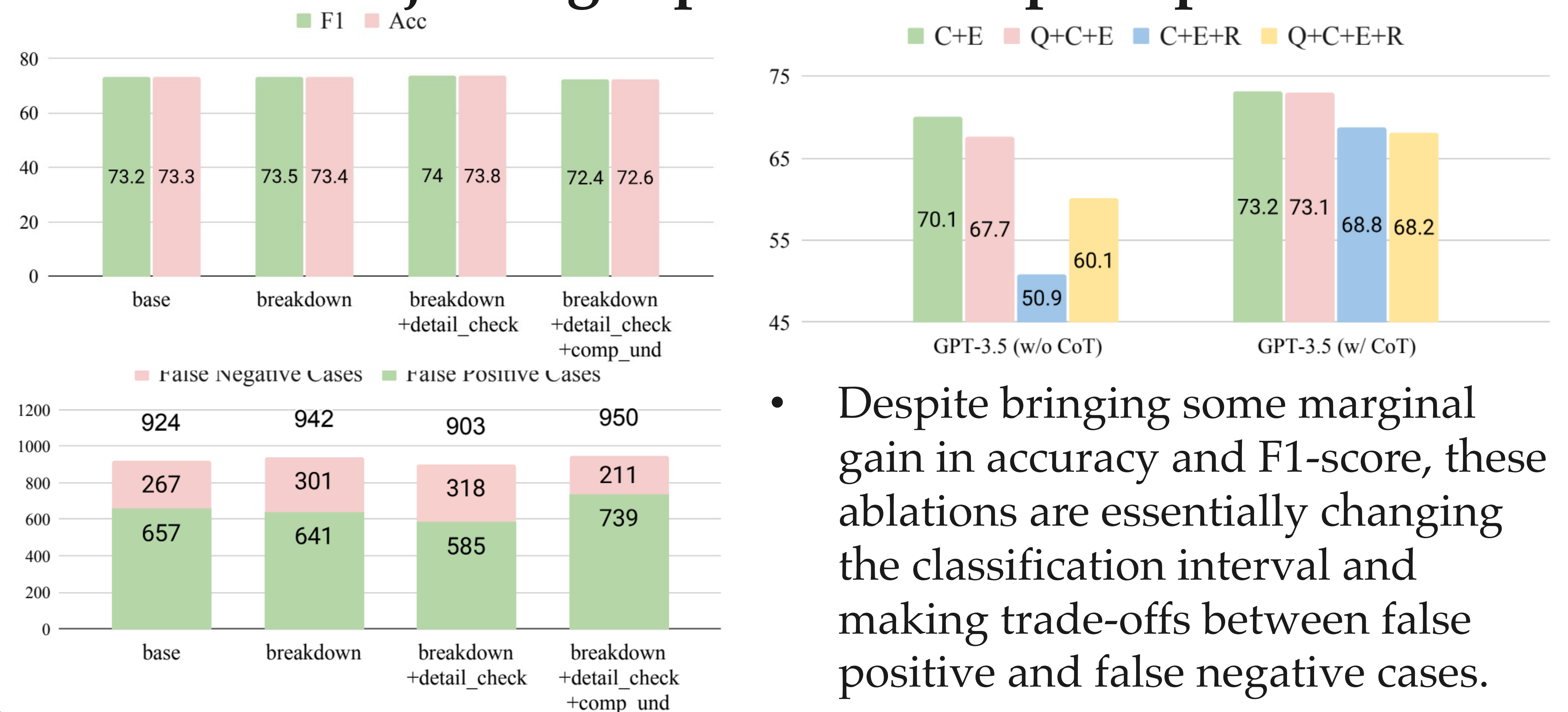
Is automatic attribution evaluation ready for practical use? **Not really!**

- Fine-tuning does provide +9.0% / +4.9% gain on macro-F1 score for ID / OOD
- However, even fine-tuned GPT-3.5 achieves a <80% F1 score

Benchmark Statistics

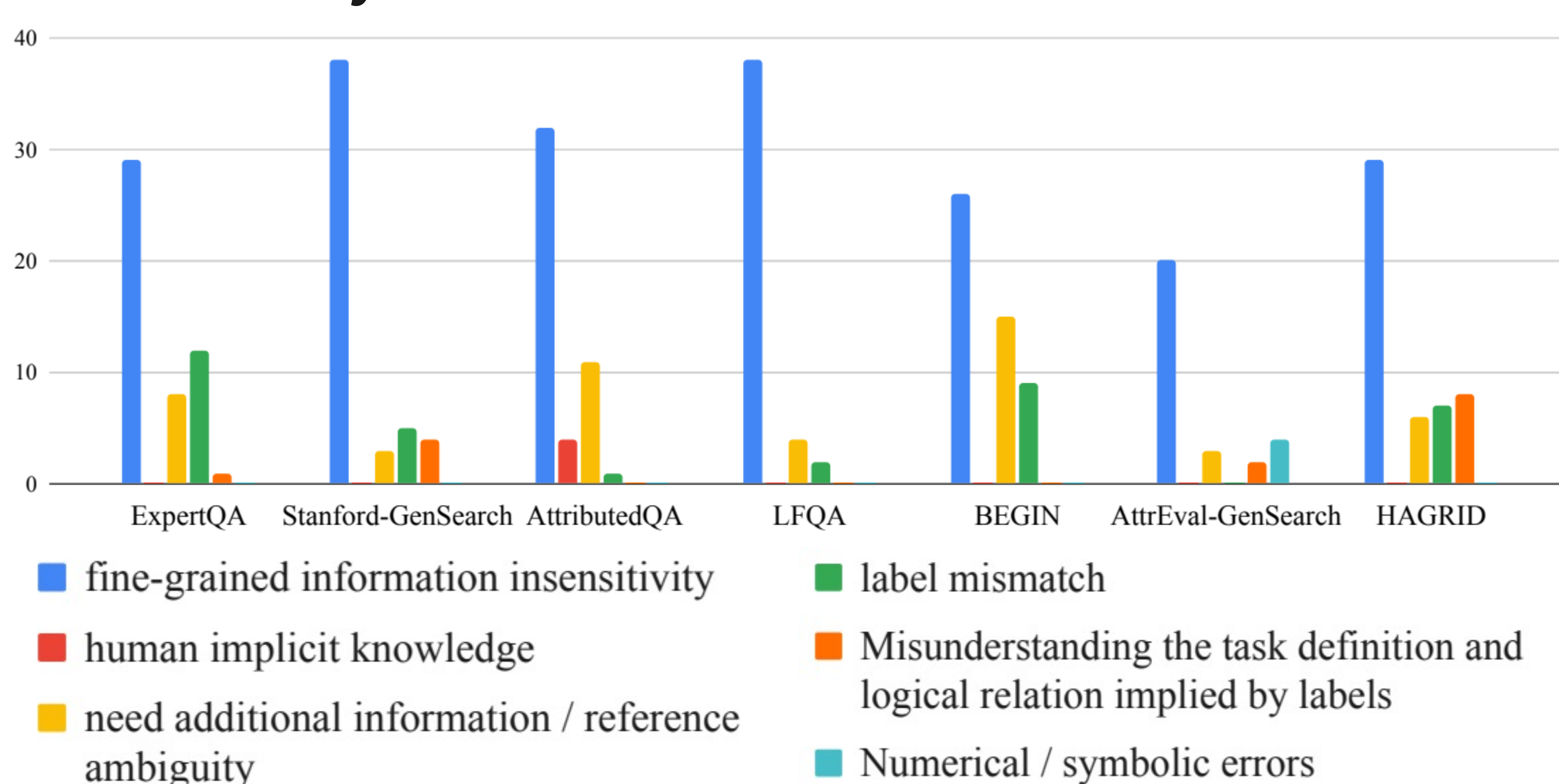
Dataset	Question Sources	#Train	#Dev	#Test
ExpertQA	Curated by domain experts	4442	470	612
Stanford-GenSearch	AllSouls, davinci-debate, ELI5, WikiHow-Keywods, NaturalQuestions	4784	544	600
AttributedQA	NaturalQuestions	2000	74	230
LFQA	ELI5	2096	110	168
ID-Total		13322	1198	1610
BEGIN	Wizard-of-Wikipedia, CMU-DoG, TopicalChat	-	-	436
HAGRID	MIRACL	-	-	1088
AttrEval-GenSearch	Curated by human annotators	-	-	162
OOD-Total		-	-	1686

Ablations: Adjusting input fields & prompts



- Despite bringing some marginal gain in accuracy and F1-score, these ablations are essentially changing the classification interval and making trade-offs between false positive and false negative cases.

Error analysis



Conclusions & Takeaways

- Systematic results show that automatically evaluating attribution is still very challenging.
- The ability of capturing fine-grained information is crucial for LLMs as good attribution evaluators.
- Multimodal grounding** should be useful for models to better utilize the information within the webpage.